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Explainability in Medical Prediction Models

הערכת עקביות ואמון בטכניקות הסבר לחיזוי תמותה בטיפול נמרץ

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Abstract

In recent years, Artificial Intelligence (AI) has become a central tool in decision-making across critical domains such as healthcare, transportation, and finance. As machine learning capabilities advance, growing concerns have emerged regarding the transparency of these models — that is, how the decisions made by the system can be understood. The field of Explainable AI (XAI) aims to address this challenge by developing methods that explain the behavior of complex models.

In high-stakes systems, such as ICU mortality prediction models, not only is the accuracy of the prediction important, but so is understanding the underlying factors that led to it. This understanding is crucial for medical teams seeking to validate or question the model's conclusions and base treatment decisions accordingly. While various explanation methods exist (e.g., SHAP, LIME, and others), there is currently no consistent scientific tool that evaluates the level of agreement or consistency between them.

This project introduces an analytical system designed to evaluate and compare various explanation methods for machine learning models. The system processes explanation outputs from multiple techniques (SHAP, LIME, and Inherent) and computes quantitative metrics such as Kendall's W, Pearson Correlation, Agreement on Direction, and Intersection@K across hundreds of different predictions. The output includes statistical analyses and interactive visualizations that demonstrate the degree of agreement between the explanation methods.

תקציר

בשנים האחרונות, הבינה המלאכותית (AI) הפכה לכלי מרכזי בקבלת החלטות בתחומים קריטיים כגון רפואה, תחבורה וכלכלה. עם ההתקדמות ביכולות הלמידה החישובית, עלו חששות גוברים בנוגע לשקיפות המודלים – כלומר, היכולת להבין את ההחלטות שהמערכת מקבלת. תחום הבינה המלאכותית ברת ההסבר (XAI) שואף לתת מענה לחשש זה באמצעות פיתוח שיטות שמסבירות את פעולת המודלים.

במערכות קריטיות כמו חיזוי תמותה בטיפול נמרץ (ICU), לא די בתחזית מדויקת – יש צורך בהבנת הגורמים שהובילו אליה. הבנה זו חיונית לצוותים רפואיים לצורך אימות או ערעור על מסקנות המודל. אף על פי שקיימות כיום שיטות הסבר שונות (כגון SHAP, LIME ואחרות), אין כיום כלי מדעי עקבי להערכת רמת ההסכמה והעקביות ביניהן.

הפרויקט הנוכחי מציע מערכת אנליטית להשוואה כמותית בין שיטות הסבר שונות. המערכת מפיקה מדדים סטטיסטיים (Kendall's W, מתאם פירסון, Agreement on Direction ו־Intersection@K) על בסיס מאות תחזיות, ומציגה תובנות באופן חזותי אינטראקטיבי. מטרת המחקר היא לבחון את מידת ההסכמה בין שיטות ההסבר, תוך שאיפה לשפר את אמון המשתמשים ולהנגיש את התובנות למשתמשים קליניים.

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1 Introduction

1.1 Background

In recent years, Artificial Intelligence (AI) has become a central tool in decision-making across critical domains such as healthcare, transportation, and finance. As machine learning capabilities advance, growing concerns have emerged regarding the transparency of these models — that is, how the decisions made by the system can be understood. The field of Explainable AI (XAI) aims to address this challenge by developing methods that explain the behavior of complex models.

1.2 Motivation

Understanding the reasoning behind model predictions is just as important as the predictions themselves, particularly in high-stakes domains such as ICU and hospital-wide mortality prediction. For clinicians, the ability to interpret a model's output enables critical assessment of its reliability and supports more informed treatment decisions. While several explainability techniques have been developed — including SHAP, LIME, and others — they often highlight different features or yield conflicting explanations for the same prediction. Currently, there is no standardized scientific framework to evaluate the consistency or agreement between these methods. This project aims to address that gap by systematically analyzing and comparing explanation outputs across techniques.

1.3 Current Landscape

Many of the existing explanation techniques differ not only in the scope of their outputs — with some providing local interpretations (focused on individual predictions) and others offering global insights into the model as a whole — but also in their underlying assumptions, algorithms, and approximation strategies. However, systematically comparing these methods is challenging, especially when multiple explanations for the same model may lead to different or even conflicting conclusions. At present, there is no widely accepted research standard that offers agreed-upon metrics for comparing explanation methods.

1.4 Research Gap

Although tools exist for interpreting model decisions, there is no clear infrastructure or structured methodology for comparing the consistency between different explanation techniques — that is, to what extent they lead to similar conclusions when evaluating the same predictions. A mathematical and empirical analysis that measures the correlation between different explanation methods and provides reliable insights into their relative trustworthiness is lacking.

1.5 Proposed Solution

This project introduces an analytical system designed to evaluate and compare various explanation methods for machine learning models. The system processes explanation outputs from multiple techniques (SHAP, LIME, and Inherent) and computes quantitative metrics such as Kendall's W, Pearson Correlation, Agreement on Direction, and Intersection@K across hundreds of different predictions. The output includes statistical analyses and interactive visualizations that demonstrate the degree of agreement between the explanation methods across both ICU-specific and general hospital datasets.

2 Literature Survey

2.1 Executive Summary

This literature survey focuses on recent research in explainability techniques applied to clinical machine learning, with an emphasis on methods such as SHAP, LIME, and model-inherent approaches. It also examines studies comparing local and global interpretability, methods for quantifying explanation consistency, and the challenges of applying explainability in real-world healthcare settings. The goal is to highlight existing approaches, identify their strengths and limitations, and position the current study within this evolving landscape.

2.2 Background Review

Explainable Artificial Intelligence (XAI) has emerged as a critical area of research, especially in healthcare, where transparency and trust are essential. AI models are increasingly used for analyzing medical data, but complex models, such as deep neural networks or ensemble methods, often function as "black boxes," making it difficult to interpret how decisions are made [1].

2.2.1 The Need for Explainability in Healthcare

In healthcare, transparency is crucial for accurate and trustworthy clinical predictions. For example, in Intensive Care Units (ICUs), predicting mortality or readmission requires models that not only produce accurate results but also provide clear explanations for clinicians. Systems that offer insight into their decision-making processes help build trust and reduce errors [2].

2.2.2 Core XAI Approaches

XAI techniques can be broadly classified into two categories:

- **Local Explainability:** Focuses on explaining individual predictions. For example, LIME (Local Interpretable Model-agnostic Explanations) generates simplified models to explain specific predictions, providing localized insights into the model's behavior.

- **Global Explainability:** Provides an overall understanding of the model's behavior across the dataset. For instance, regression coefficients in logistic regression models offer a straightforward way to interpret feature importance at the global level. Other methods, such as SHAP (SHapley Additive exPlanations), excel in providing both global and local insights by attributing importance scores to features across all predictions while remaining interpretable at the instance level [1, 3].

2.3 Relevant Literature Review

Explainable AI (XAI) has been widely adopted in healthcare for improving transparency and trust in machine learning models. The following section summarizes key studies that align with the objectives of this research, focusing on explainability techniques, their applications, and challenges.

2.3.1 Explainability Techniques in Machine Learning

LIME and SHAP for Transparent Health Predictions: Ashraf et al. (2024) examine the usability of LIME and SHAP for providing explainability in healthcare settings [4]. They highlight that SHAP excels in global feature analysis while LIME is effective for local interpretations. However, inconsistencies in feature importance rankings across techniques underline the need for consistency evaluation.

Approximating XGBoost with an Interpretable Decision Tree: Sagi and Rokach (2021) introduce a method to approximate complex gradient-boosted decision trees (GBDTs) with a single interpretable decision tree [5]. While this approach balances the trade-off between predictive accuracy and interpretability, it represents one of many techniques aimed at improving transparency in machine learning models. This method was initially considered during the project, but ultimately not adopted.

2.3.2 Applications in ICU and Healthcare Settings

Improving ICU Alarm Systems with SHAP: González-Nóvoa et al. (2021) apply SHAP to improve ICU alarm systems by identifying critical features for mortality predictions [6]. Their study shows that explainability techniques can reduce false alarms and enhance clinicians' trust in automated systems.

Comparative Analysis of Explainable Models: Stenwig et al. (2022) conduct a comparative analysis of explainable models for predicting hospital mortality [7]. Their work evaluates models such as random forests and adaptive boosting using SHAP, emphasizing variability in outputs.

2.3.3 Evaluating Consistency and Metrics

Local vs. Global Interpretability: Kopitar et al. (2019) compare local and global interpretability approaches in the context of Type 2 Diabetes Mellitus screening, showing how combining these methods yields a more comprehensive understanding [8].

Metrics for Evaluating Explainability: Muñoz et al. (2023) introduce quantitative evaluation metrics for assessing the reliability of explainability techniques [9]. These metrics provide a foundation for comparing methods like SHAP, LIME, and interpretable decision trees.

2.3.4 Research Gaps and Contributions

The literature reveals several gaps that this research aims to address:

- **Inconsistencies in Explainability Outputs:** Methods like SHAP and LIME often provide conflicting feature importance rankings [4, 7].
- **Lack of Unified Tools:** Limited integration of local and global explainability approaches [8].
- **Need for Interactive Visualizations:** A lack of interactive platforms to compare outputs of different explanation methods [6].

This research addresses these gaps by developing an interactive visualization tool that integrates multiple explainability techniques, enabling clinicians to access global and local insights while improving trust and usability.

3 Research Goals

The overarching goal of this study is to examine the level of consistency and agreement between different explanation methods for machine learning models, with a focus on critical medical scenarios, including ICU mortality prediction and general inpatient outcomes.

To achieve this, the study aims to accomplish the following objectives:

- **Develop an analytical infrastructure** that enables systematic comparison between commonly used explanation methods (SHAP, LIME, Inherent) across a large number of instances.
- **Provide a practical tool** for researchers and professionals to make informed decisions when selecting an explanation method based on its specific application context.

4 The research methodology

The study followed a structured and quantitative process for comparing explanation methods for machine learning models, integrating statistical tools, visualization techniques, and interactive analysis. The methodology included the following stages:

Data Collection and Processing

Two separate medical datasets were used, each containing clinical data related to hospitalized patients, although only one of them specifically focuses on Intensive Care Unit (ICU) cases.

The first dataset, GOSSIS (Global Open Source Severity of Illness Score), was obtained from the WiDS Datathon 2020 and includes detailed ICU records from multiple hospitals worldwide. It provides rich information about critically ill patients, including vital signs, lab tests, interventions, and mortality outcomes. The dataset focuses on clinical data collected during the last 24 hours of ICU stays.

The second dataset, the Diabetes 130-US hospitals for years 1999–2008, was retrieved from the UCI Machine Learning Repository. This dataset contains records of over 100,000 diabetic patients hospitalized across the U.S. It is not ICU-specific but includes various hospitalization encounters and relevant clinical features such as diagnoses, procedures, and readmission data. Although this dataset does not reflect ICU settings, it was chosen to test the generalizability of our consistency metrics across different medical contexts.

Each dataset underwent a process of cleaning, filtering, and normalization to enable consistent comparison between explanation methods. Working with two different datasets allowed for evaluating whether the insights obtained were consistent across varying data sources.

Model Training

Several machine learning models were trained to predict patient mortality, including:

- XGBoost
- Logistic Regression
- Decision Tree

Each model was applied to the prepared data, and predictions were analyzed individually to generate explanations.

Generating Explanations

For each prediction, explanations were generated using three primary methods:

- SHAP
- LIME
- Inherent — a built-in explanation unique to the model itself, requiring no external method.

Each method produced a ranking of feature contributions to the prediction.

Explanation Methods Overview

The three explanation techniques used in this project are as follows:

1. **SHAP (SHapley Additive exPlanations)**:
   A model-agnostic method based on cooperative game theory that assigns each feature

an importance value for a particular prediction. SHAP provides both global and local explanations.

2. **LIME (Local Interpretable Model-agnostic Explanations)**:
   Another model-agnostic method that focuses on explaining individual predictions by training an interpretable surrogate model (typically linear) on perturbed samples near the input instance. LIME offers local interpretability and helps identify which features had the most influence on a specific prediction. For the global explanation, an average was taken across 1,500 patients, with a local explanation generated for each of them.

3. **Inherent Explanations**:
   These are explanations derived directly from the internal structure and parameters of each model, without applying external post-hoc interpretation techniques. The explanation methods are divided into **local** (per-instance) and **global** (dataset-level) approaches, as follows:

   - **Logistic Regression**:
     - *Local:* Contributions are computed per instance by scaling each feature's coefficient using the derivative of the sigmoid function:

       $$c_i = \sigma'(z) \cdot w_i \cdot x_i, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)), \quad z = \mathbf{w}^\top \mathbf{x} + b$$

       This captures the local non-linearity around the prediction point.
     - *Global:* Feature importance is based on the absolute value of the learned coefficients:

       $$\text{Importance}(x_i) = |w_i|$$

       representing the strength of each feature's influence across the dataset.
   - **Decision Tree**:
     - *Local:* The instance's path through the tree is traced, and feature contributions are measured by the change in predicted probability at each node:

       $$\Delta p = p_{\text{child}} - p_{\text{parent}}$$

     - *Global:* Feature importance is computed via impurity reduction (e.g., Gini) across all splits:

       $$\text{Importance}(x_i) = \sum_{\text{splits on } x_i} \Delta\text{Impurity}$$

   - **XGBoost**:
     - *Local:* Uses the built-in `pred_contribs` method to decompose predictions into additive contributions from each feature and bias:

       $$\hat{y} = \sum_{i=1}^{n} c_i + \text{bias}$$

– *Global:* Feature importance is based on the `gain` metric, reflecting average loss reduction from each feature's splits:

$$\text{Gain}(x_i) = \frac{1}{N_i} \sum_{\text{splits on } x_i} \Delta\text{Loss}$$

Quantitative Consistency Metrics

The project involved statistical calculations between pairs of explanation methods (e.g., SHAP, LIME, Inherent) to assess the degree of consistency between them. The aim was to determine whether the methods yielded similar insights when explaining the same prediction for the same model.

Each metric was calculated for every pair of methods, and the results were averaged — both at the local level (per individual prediction) and at the global level (across the entire model). These averages reflect the overall consistency between the methods across all tested predictions.

The following quantitative metrics were used:

- **Kendall's W:** A ranking consistency measure. This metric evaluates how much agreement exists between explanation methods regarding the ranking of features by importance. Each method ranks features, and Kendall's W (ranging from 0 to 1) measures the level of agreement between these rankings.
- **Pearson Correlation:** A linear correlation metric between feature importance values. It measures the numerical similarity between the importance scores assigned by different methods by computing the linear correlation (straight-line relationship) between two value vectors, one from each method.
- **Agreement on Direction:** Measures agreement on the direction of impact (used only in local explanations). This metric is relevant for local explanations, where a feature can contribute positively or negatively to a prediction. It checks whether the methods agree on the direction of influence.
- **Intersection@K:** Compares the top-K most important features identified by each explanation. The metric checks how many of the top-K features (e.g., K=5) appear in both methods' outputs.

Development of an Interactive Interface for Presenting Findings

Global Level: Displays global explanations for each method, with a variety of chart types and user-friendly layouts.

Local Level: Shows detailed explanations for individual patient predictions based on the selected dataset. The dashboard enables navigation between samples, filtering by model/method/metric, and a range of visual comparisons between explanation methods.

Analysis of Results

Patterns of agreement and disagreement between the methods were examined, as well as whether certain explanation methods demonstrated greater consistency across models and datasets. A comparison between the datasets was also conducted to identify recurring patterns, differences, or unique insights tied to each data source.

5 Interface Design

5.1 Main Screen

Navigation menu - global and local explanations

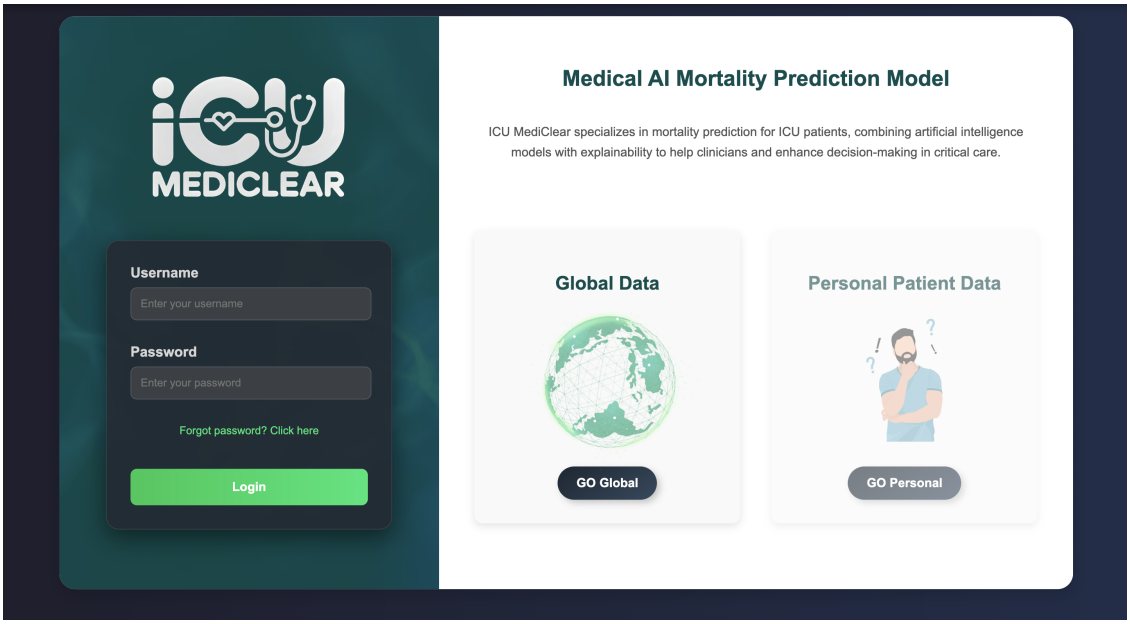


Figure 1: Main Landing Page of the System

This screen represents the system's opening view when no user is currently logged in. In this state, access to individual patient data (from either dataset) is restricted. However, the public global dashboard remains fully accessible to all visitors.

It is important to note that, for security and medical confidentiality reasons, self-registration is not permitted. Only users with an Admin role have the authority to register new users in an organized and approved manner. Admin users can also delete accounts or temporarily suspend user access (for cases where temporary access restriction is required without permanently deleting the account).

User Dashboard (Admin View)

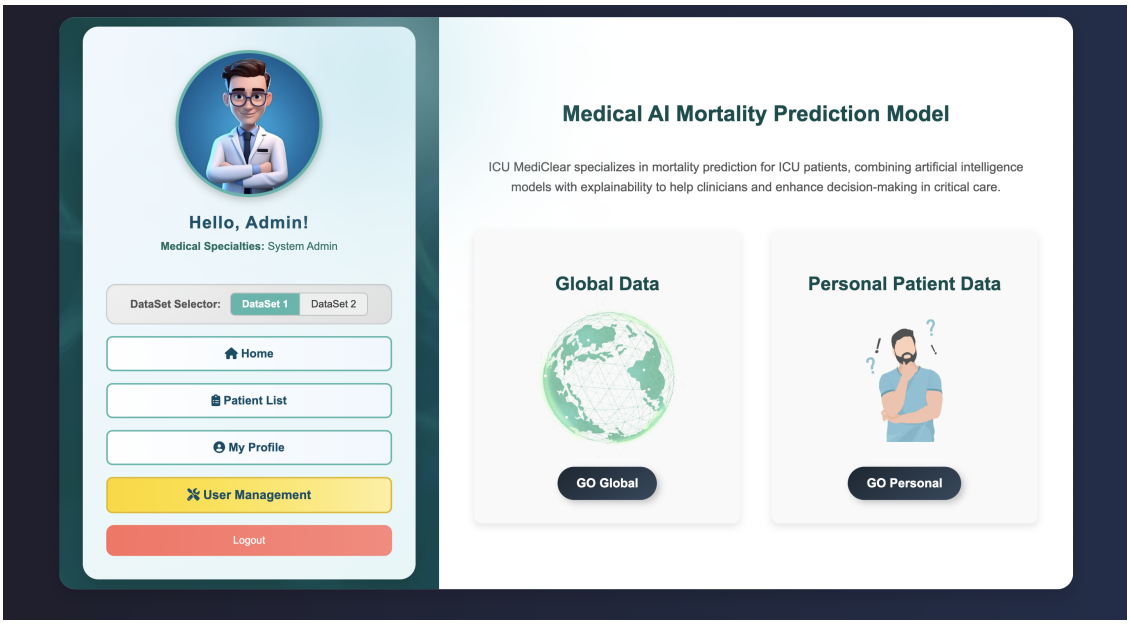


Figure 2: Screen Displayed After Logging in with Username and Password

This view appears once a user successfully logs into the system using valid credentials. Inactive users cannot log into the system — this likely indicates that their access has been suspended by an Admin.

Patient List View

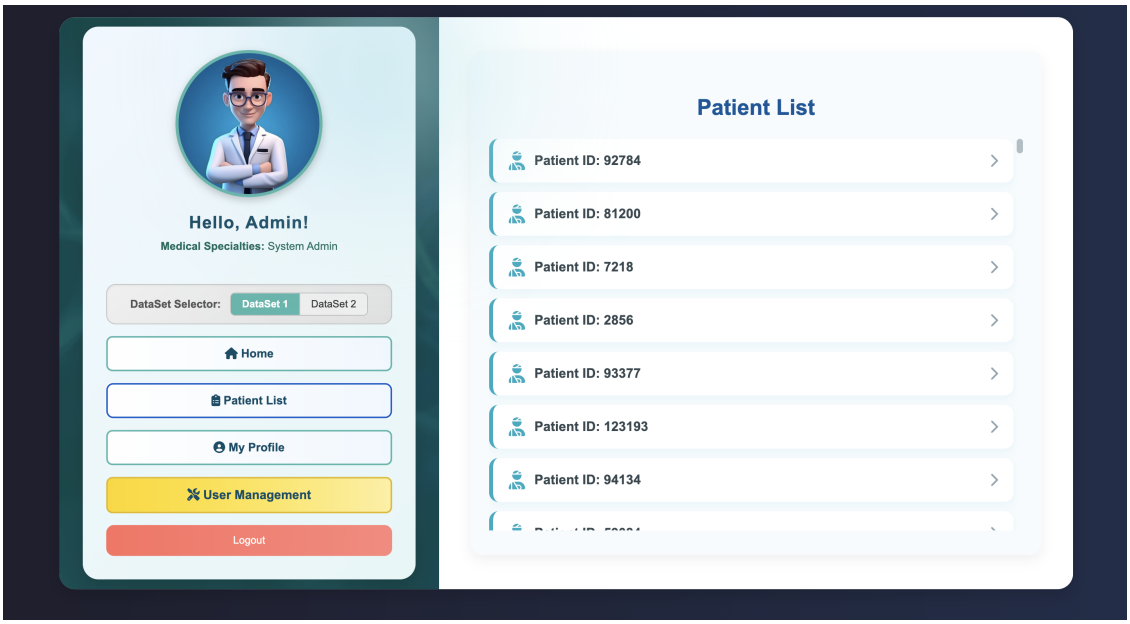


Figure 3: Patient List View (According to the Selected Dataset)

This window displays a list of patients based on the currently selected dataset. Clicking on any patient will navigate to the local explanation page specific to that patient.

User Profile View

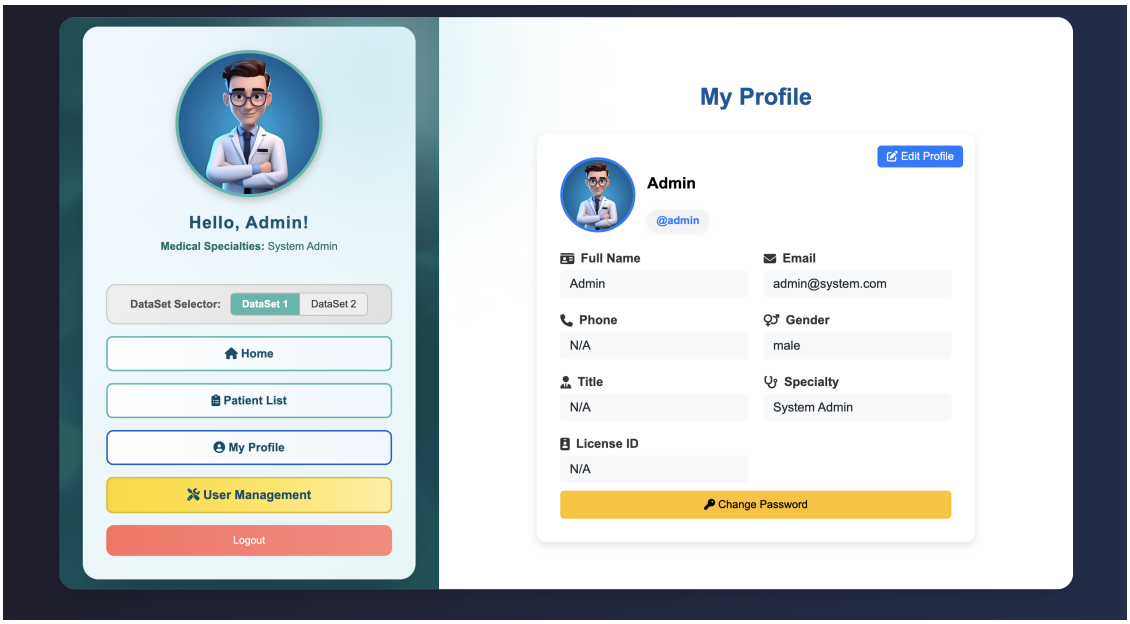


Figure 4: User Profile View

This window allows users to update their personal information and change their password. Admin users have extended privileges — they can reset passwords and update profile details for any user in the system. In contrast, regular users have limited access and can only update a restricted set of personal details.

Admin Panel - User Management Dashboard

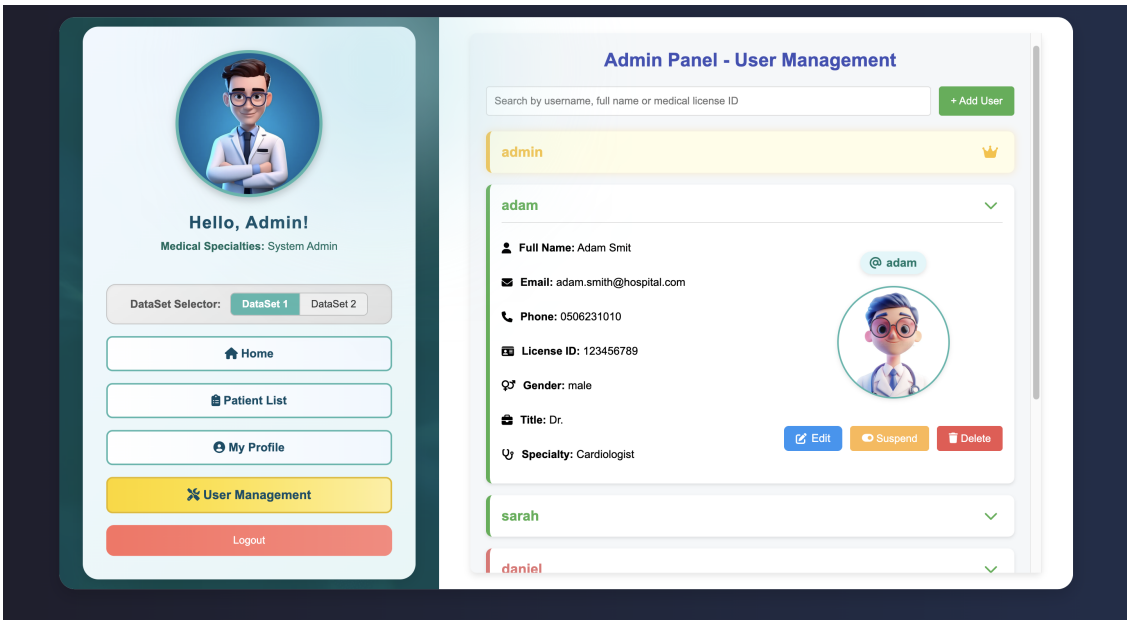


Figure 5: Figure 5: User Management Dashboard

This view is accessible **only to Admin users**. It displays a list of all registered users in the system.

- A user highlighted in green is active and has full access to the system's data.
- A user highlighted in red is currently **blocked** from accessing the system.

Admin users have full control over user management, including:

- Adding new users
- Deleting users
- Suspending or reactivating suspended users
- Resetting user passwords
- Editing user details

5.2 Global View

Graphs and feature importance maps (e.g., bar chart).

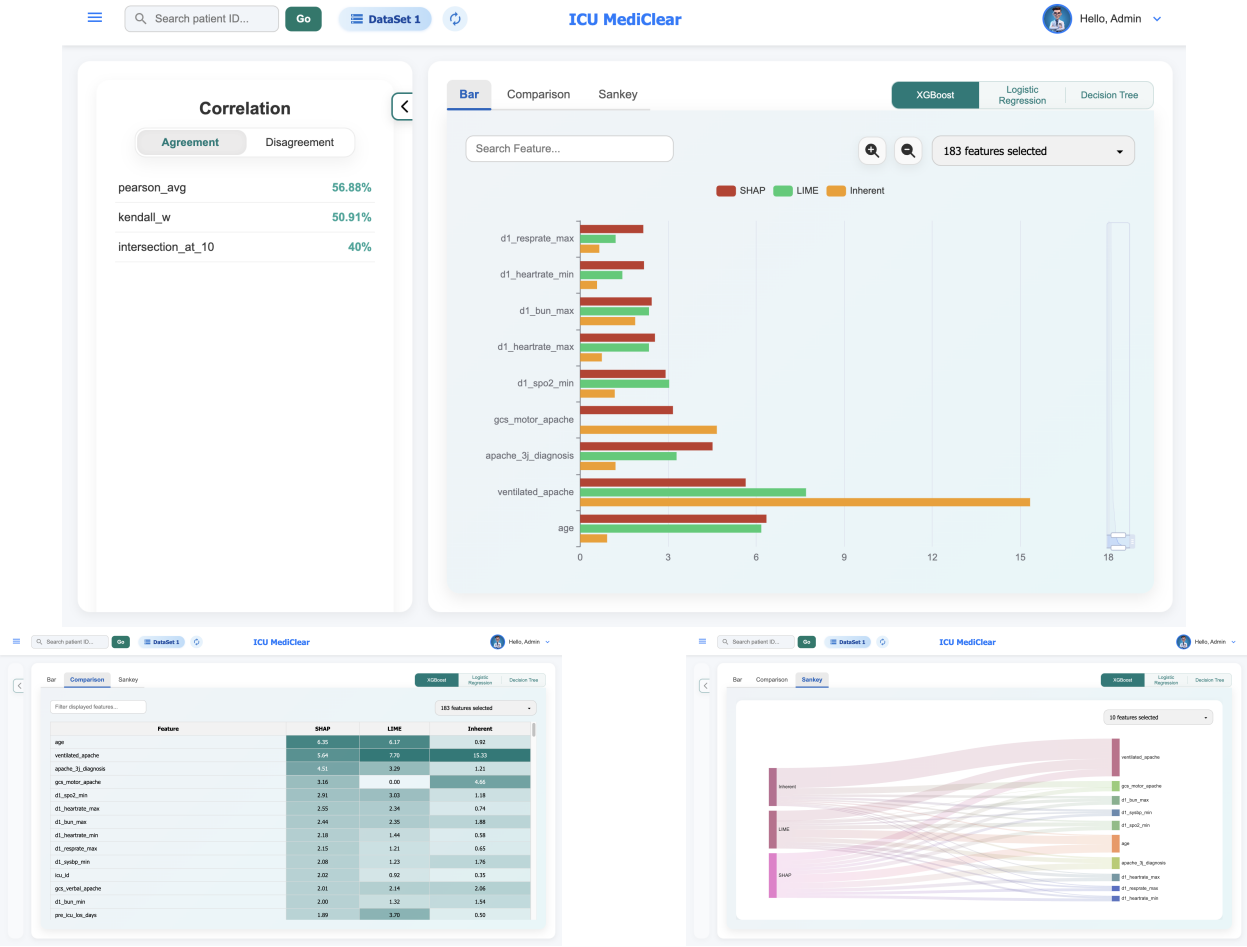


Figure 6: Global View Examples

5.3 Local View

Graph displays for each specific patient

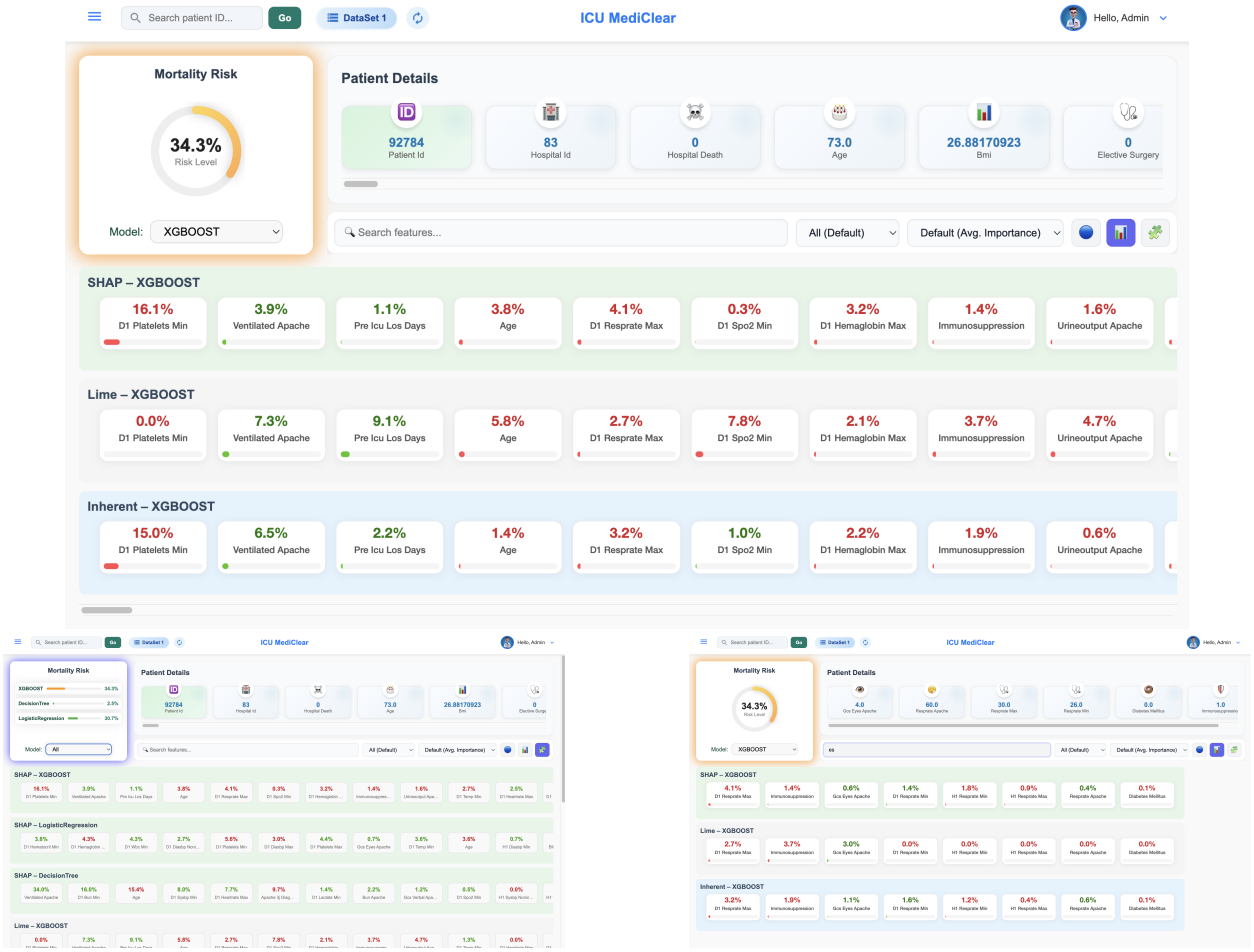


Figure 7: Local View Examples

6 The experiments

To evaluate the research objectives and assess the consistency of explanation methods, a series of controlled experiments were conducted using two separate medical datasets, each containing thousands of patient samples. The experiments were structured across several levels of analysis: by method, by model, and by patient.

Experiment 1: Generating Explanations Using Multiple Methods

For each prediction produced by the models, explanations were generated using three methods: SHAP, LIME, and Inherent.

This experiment was carried out across all three machine learning models: XGBoost, Logistic Regression, and Decision Tree.

The goal was to examine whether the three methods provided similar insights regarding the contributions of clinical features to the predictions.

Experiment 2: Measuring Consistency Between Methods

We computed statistical metrics between each pair of the three explanation methods for every prediction:

- **Kendall's W** – To assess the consistency of feature importance rankings.
- **Pearson Correlation** – To measure the linear correlation between feature contribution values.
- **Agreement on Direction** – To compare the direction of influence (positive or negative) for each feature.
- **Intersection@K** – To compare the top-K most influential features identified by each method.

The results were compiled into summary tables and used to compare between models, methods, and datasets.

Experiment 3: Dataset Comparison

All experiments were repeated using two distinct medical datasets collected from independent sources. This experiment aimed to answer the question:

> *Do the consistency metrics yield similar results when the data source changes?*

A comparative analysis of metric distributions was conducted, including graphical visualizations, mean comparisons, and standard deviation analysis.

7 Results

At the global level, explanation methods show higher agreement in simpler models (such as Logistic Regression and Decision Trees) compared to more complex models like XGBoost. There is a strong correlation between the explanation methods.

Below are the results as presented in the application we developed:

7.1 Global

- **XGBoost:**

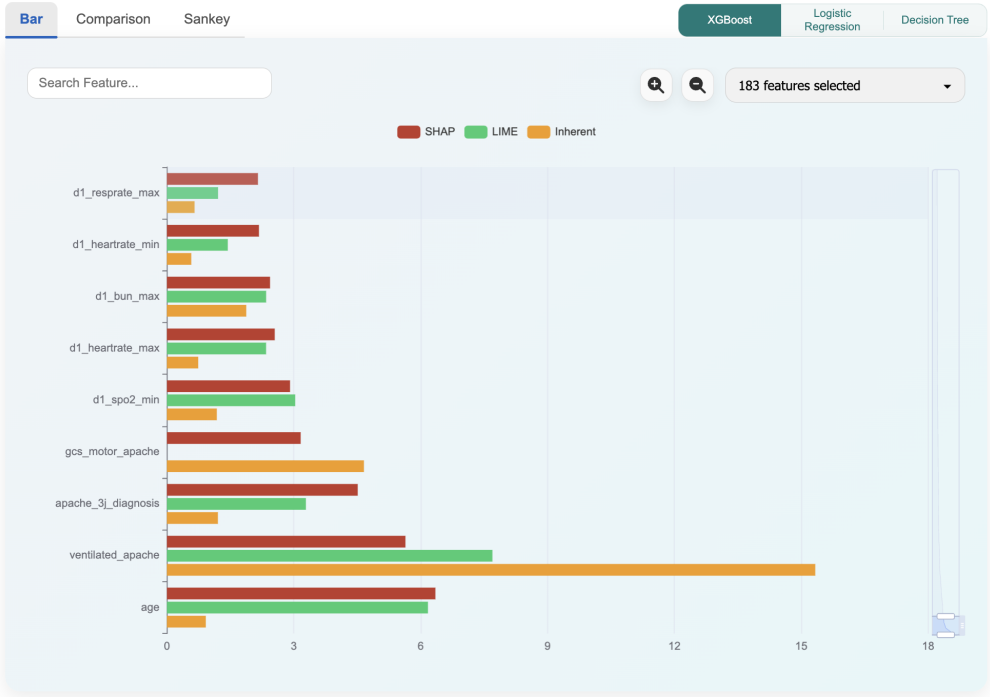


Figure 8: Global Explanation Results - XGBoost

- **Logistic Regression:**

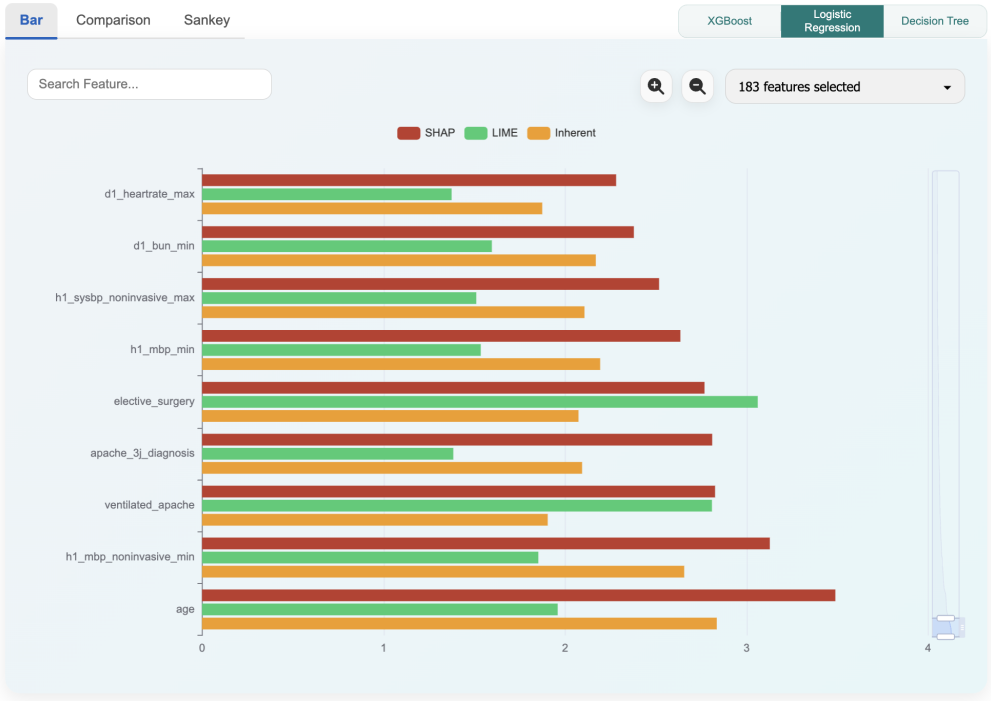


Figure 9: Global Explanation Results - Logistic Regression

- **Decision Tree:**

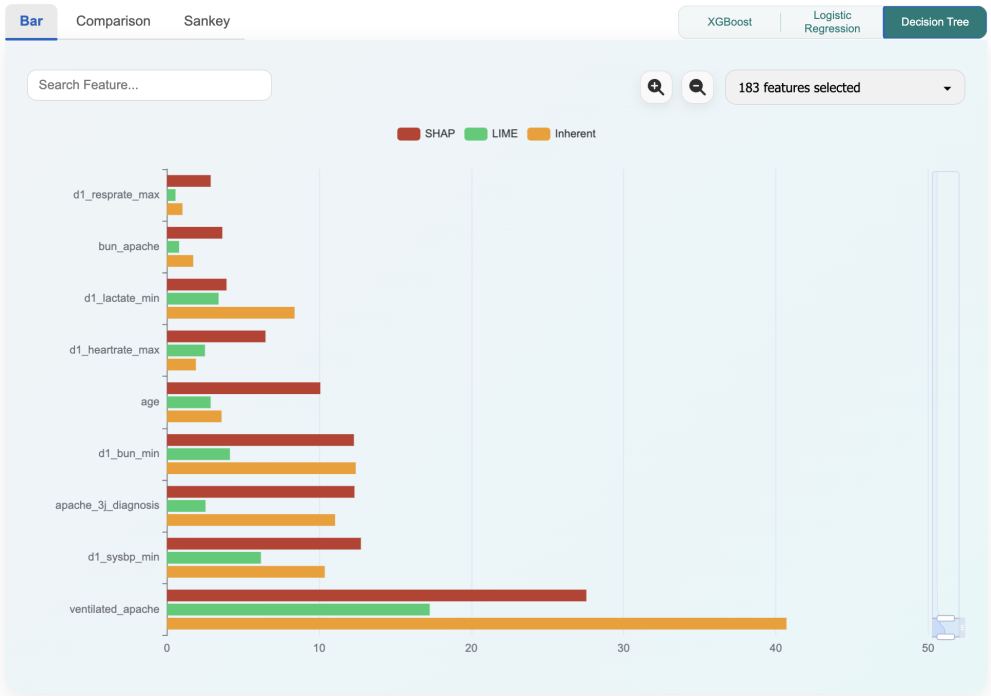


Figure 10: Global Explanation Results - Decision Tree

7.2 Local

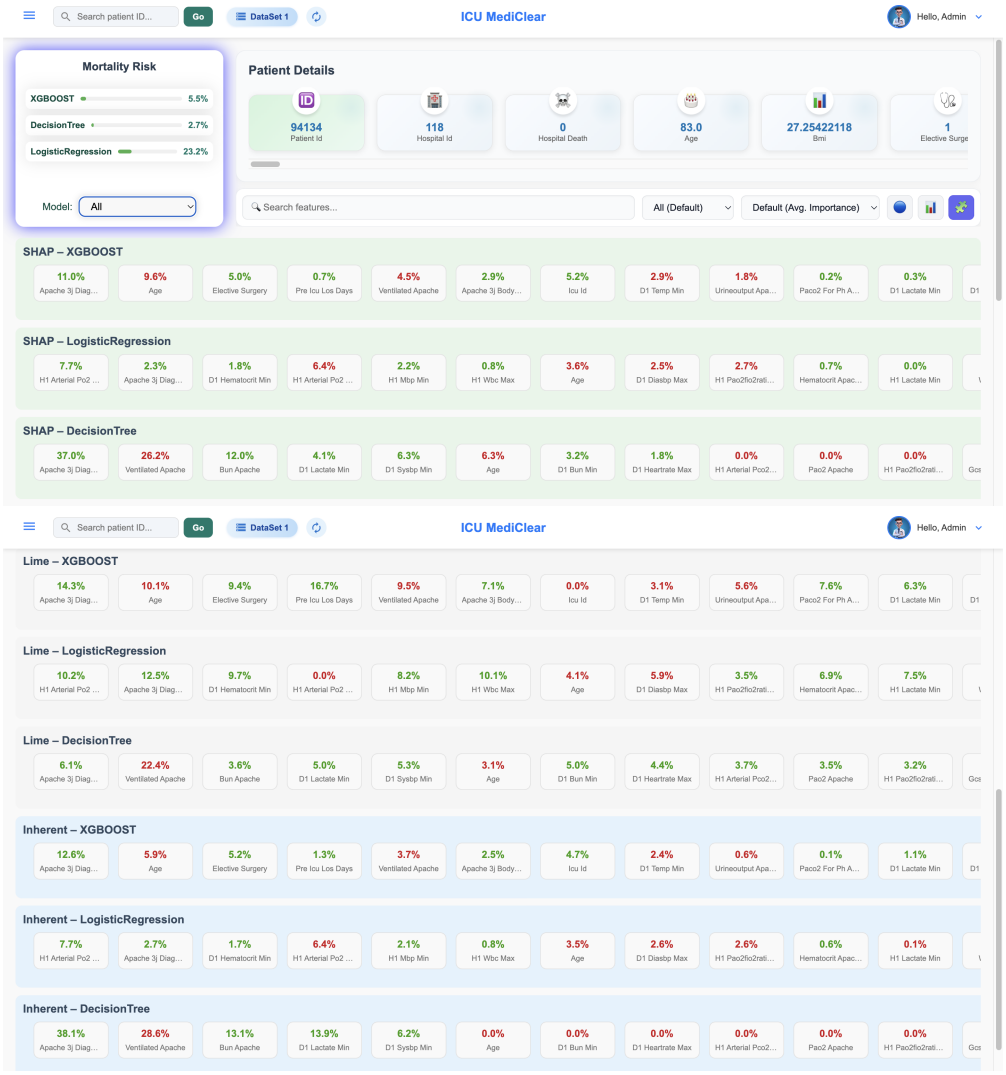


Figure 11: Local Explanation Results - XGBoost, Logistic Regression and Decision Tree

The following are the results obtained after running correlation analysis methods:

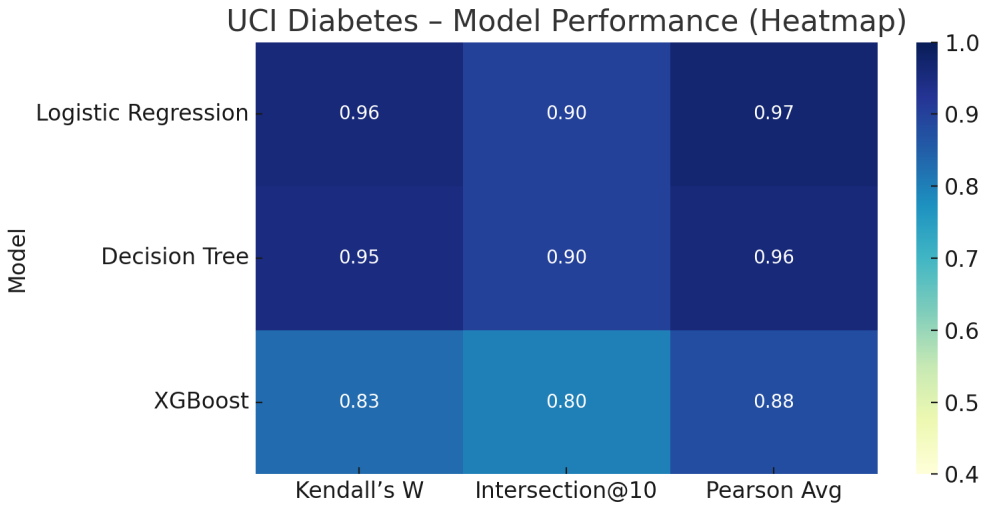


Figure 12: Global - UCI (DataSet 2 in MediClear)

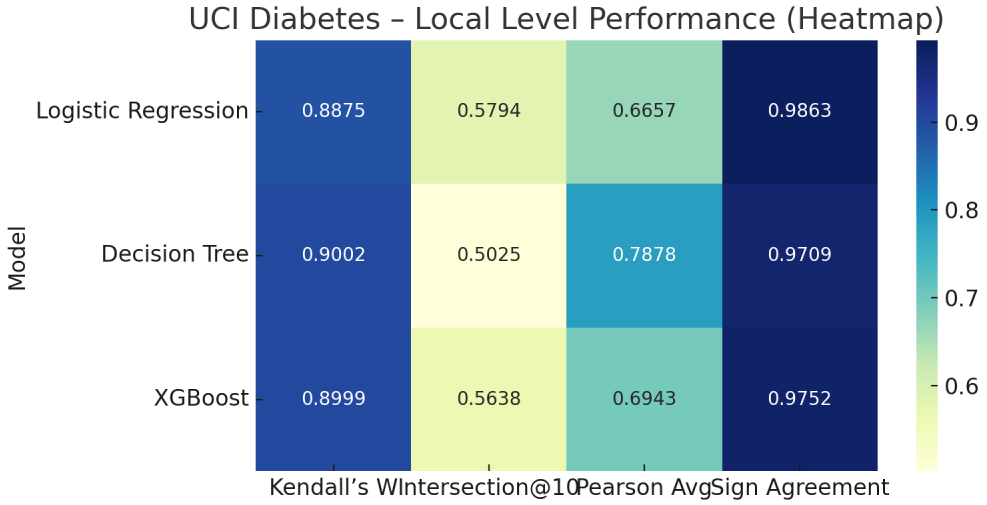


Figure 13: Local - UCI (DataSet 2 in MediClear)

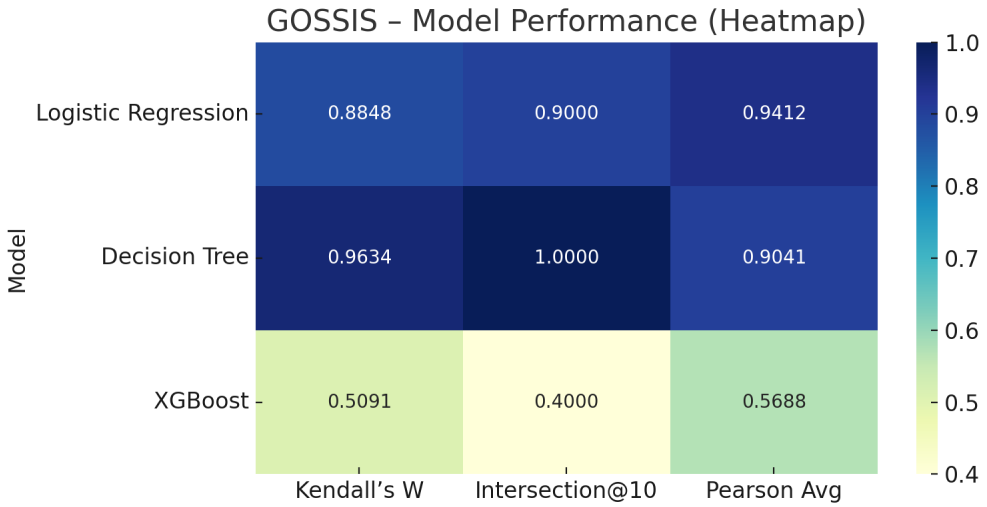


Figure 14: Global - GOSSIS (DataSet 1 in MediClear)

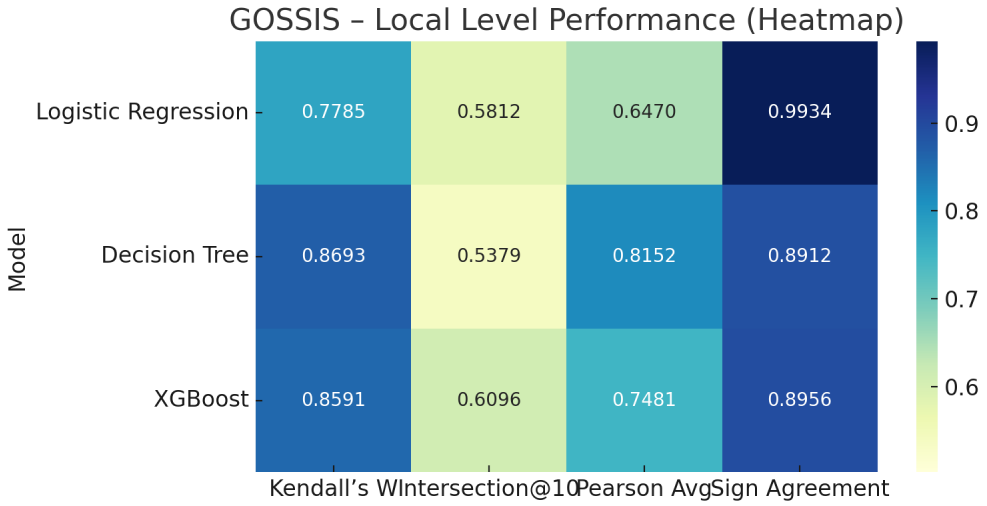


Figure 15: Local - GOSSIS (DataSet 1 in MediClear)

Consistent Global Insight: Higher Agreement in Simpler Models
Across both datasets (GOSSIS and UCI), a clear pattern emerged: simpler models demonstrated higher consistency between explanation methods. This trend was most notable in the quantitative metrics used to assess agreement—Kendall's W, Pearson Correlation, and Intersection@10.
For example, in the UCI dataset, the Logistic Regression model reached a Pearson AVG of approximately 0.97, indicating near-perfect agreement in feature rankings between methods. In the GOSSIS dataset, the same model achieved a Kendall's W of 0.94. In contrast, the more complex XGBoost model showed lower agreement, with Kendall's W values of 0.88 (UCI) and 0.56 (GOSSIS).

These findings support the conclusion that simpler models produce more stable and interpretable explanations, with greater alignment between different explanation techniques regarding the importance of individual features.

8 Conclusions

The following are the main conclusions of this study. The first point below summarizes the central finding and directly answers the research question regarding the consistency between explanation methods.

- There is a high level of agreement between explanation methods.
- At the global level, explanation methods show higher agreement in simpler models (such as *Logistic Regression* and *Decision Trees*) compared to more complex models like *XGBoost*.
- At the local level, no higher agreement was observed in simpler models compared to more complex ones.

9 Future Work

Despite the contributions of this research, **several challenges remain that require further investigation and development:**

- **Lack of Ground Truth** – Currently, there is no definitive way to determine which explanation method is the "correct" one. As a result, comparisons between methods rely solely on indirect evaluation metrics.
- **Variability Due to Preprocessing Effects** – Preprocessing steps (such as scaling or handling missing values) may influence the outcomes of explanation methods. This potential impact was not deeply explored in the current study.
- **Understanding Low Agreement Cases** – In cases where explanation methods show relatively low agreement, it is important to examine the underlying causes. One possible explanation is multicollinearity. For example, one method may identify a patient's age as the primary risk factor, while another points to Alzheimer's disease. The disagreement might stem from a strong correlation between age and Alzheimer's, making it difficult for explanation methods to isolate the dominant feature.

10 References

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System Specification – MediClear

The following sketches represent the initial structure and design of the system, created during the planning phase prior to the actual implementation of the website and interface. They served as a blueprint for building the visual and functional layout of the final system.

System Name: MediClear

Purpose:
A system for predicting mortality risk in hospitalized patients, including both ICU and general hospital populations, leveraging AI-based models with an emphasis on explainability, to support better medical decision-making.

Target Audience:

- physicians and medical staff
- Medical data scientists
- Researchers in AI and explainability

General Workflow:
The system includes two main modules:

- **Global** – Enables observation and understanding of general explanation trends across models and datasets.
- **Personal Patient** – Enables detailed analysis of an individual patient, displaying feature contributions using different explanation methods (SHAP, LIME, Inherent).

Main Screens:

Login / Home Page

- Allows existing users to log in or new users to register.
- Public access to global data is available without logging in, but patient-specific data requires authentication.
- Includes a brief introduction explaining the system's purpose.

Main Screen (Post-login)

- Displays physician details (name, medical specialty, license number).
- Provides two navigation options:
  - Patient List – View list of patients
  - Personal Area – Access to personalized features
- Access to Global and Personal data views

Global Explainability Dashboard

- Displays comparative graphs of feature contributions across different models (e.g., Logistic Regression, Decision Tree) and explanation methods (e.g., SHAP, Inherent).
- Shows agreement metrics between methods visually.

Local Explanation Screen (for a specific patient)

- Displays estimated mortality risk (in %) according to the selected model (e.g., XGBoost).
- Presents detailed patient information (blood pressure, heart rate, temperature, weight, etc.).
- Visual explanation of feature contributions from each method (SHAP, LIME, Inherent) using color-coded charts and interactive visuals.

Technologies Used:

- Frontend: Vue.js / JavaScript
- Models: XGBoost, Logistic Regression, Decision Tree
- Explanation Methods: SHAP, LIME, Inherent
- Visualization: Graphs, tables, and dynamic visualizations

Design Principles:

- Clinician-friendly – Emphasis on quick access to critical information
- High-contrast color scheme to highlight risk levels and feature impact
- Clean and consistent layout for rapid identification of key features

Usage Goals:

- Improve real-time decision-making in critical care
- Help physicians understand the factors driving mortality predictions
- Increase trust in AI models by providing transparent and interpretable explanations

Interface Design (Initial Sketches)

- **Main Screen:** Navigation menu between global explanations and local explanations.

Interface Design (Initial Sketches)

- **Main Screen:**
  - Navigation menu between global explanations and local explanations.

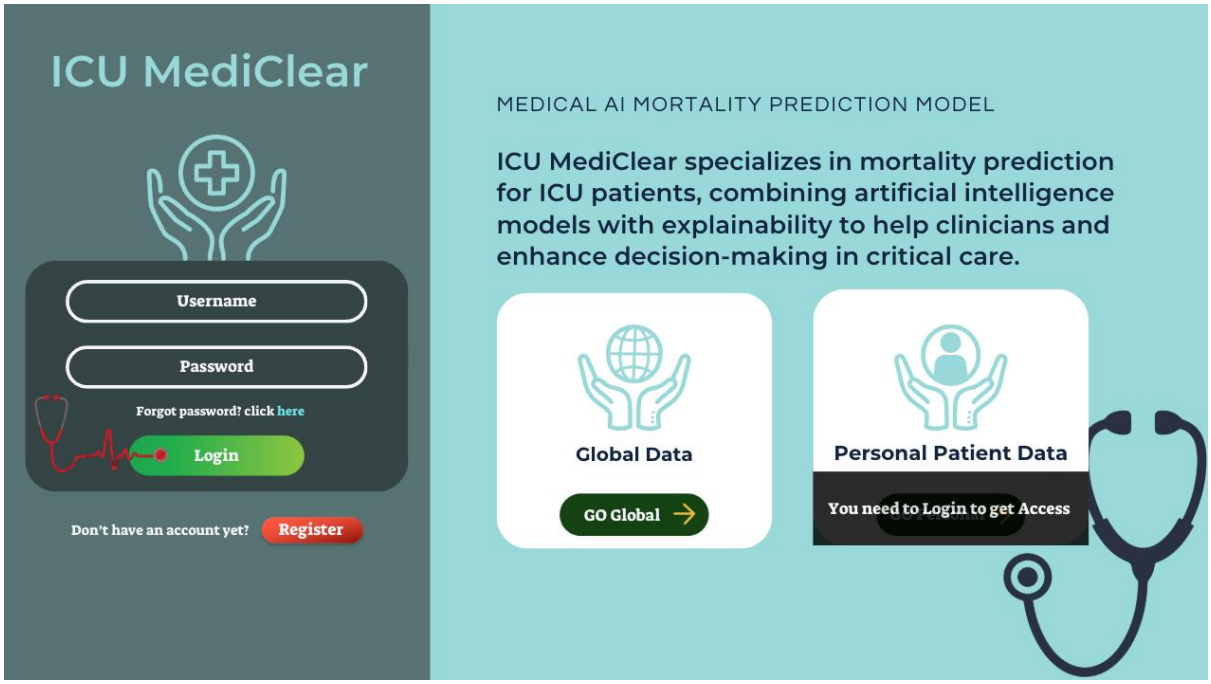


Figure 16: Main screen for a logged-out user

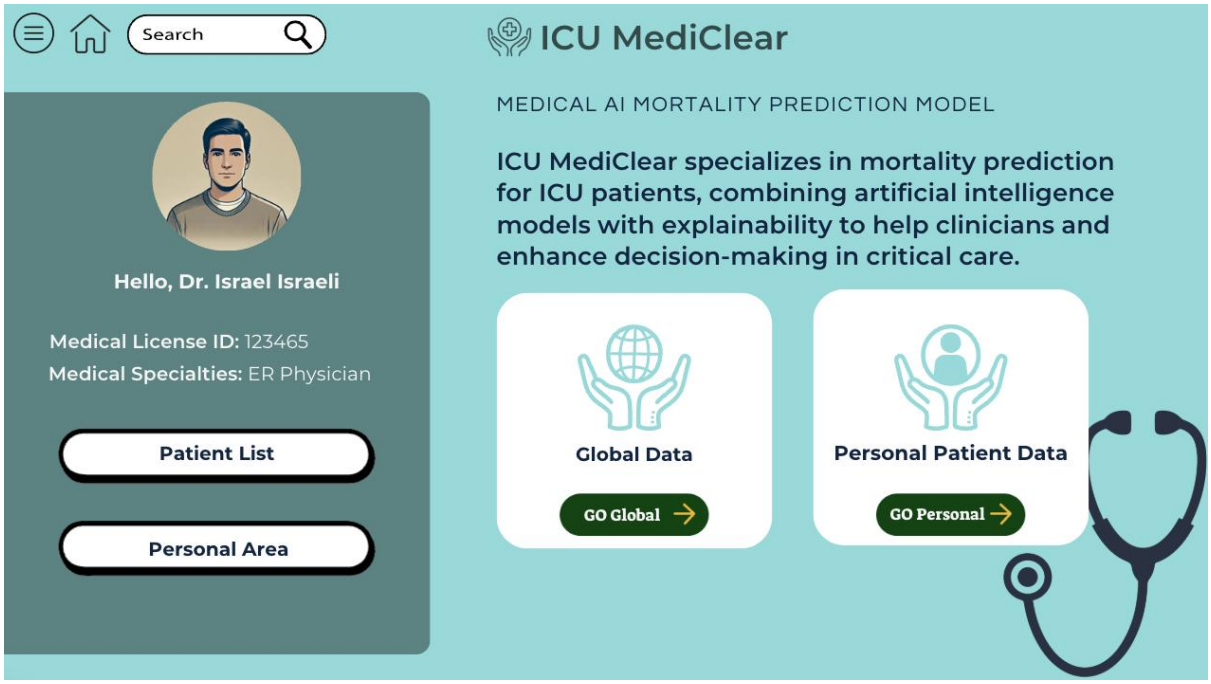


Figure 17: Main screen for a logged-in user

- **Global View:**
  - Graphs and feature importance maps (e.g., bar chart).

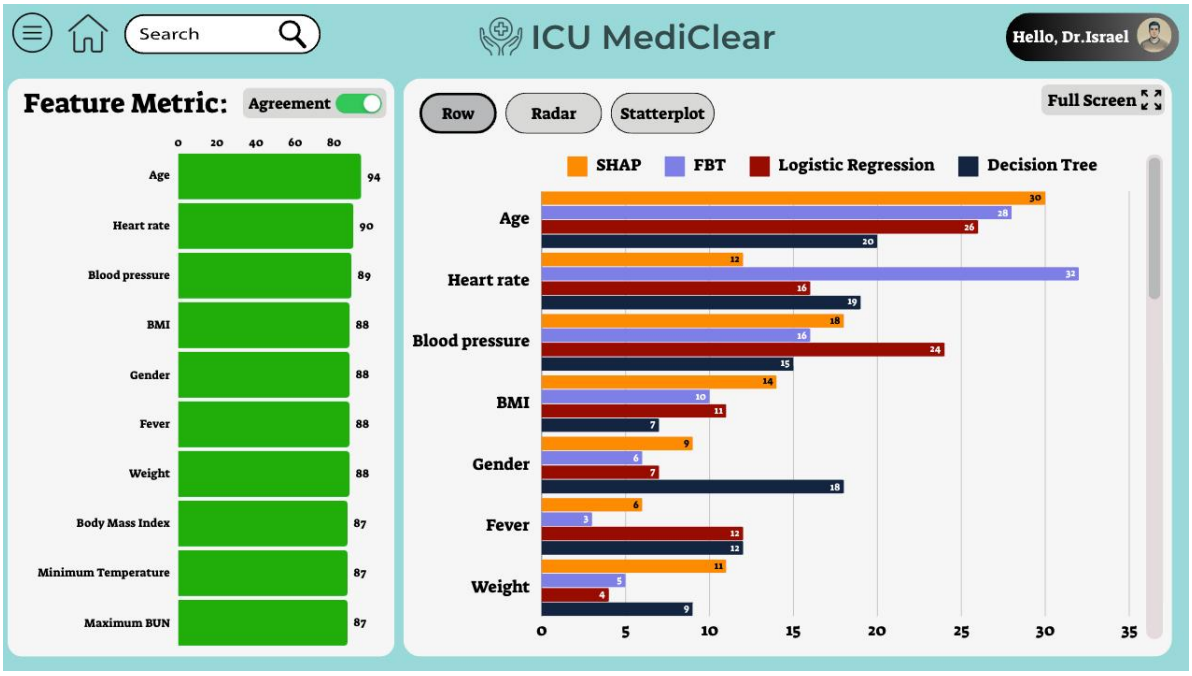


Figure 18: Global View Example

- **Local View:**
  - Graph displays for each specific patient.

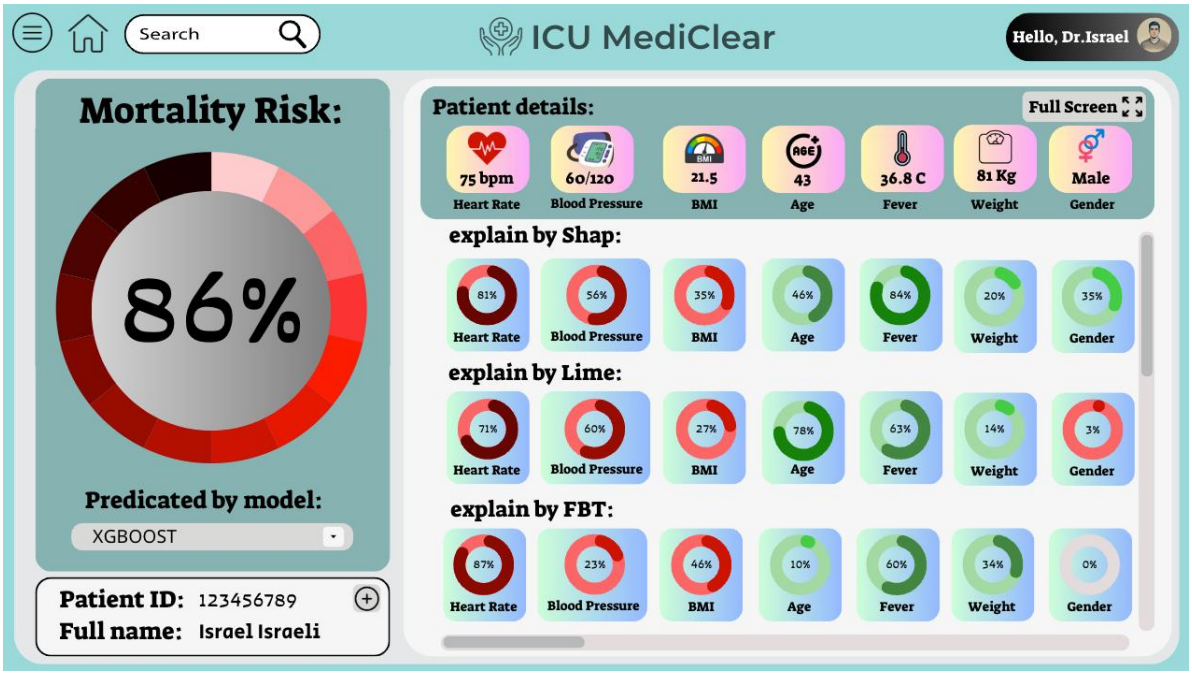


Figure 19: Local View Example